
The Coordination Deficit: Ecological Baselines and Behavioral Mechanisms for Cooperative LLM Agent Systems

Brandon Yee¹ Pairie Koh¹

Abstract

Cooperative AI requires agents that coordinate reliably at scale, yet the behavioral and ecological conditions for such coordination remain poorly understood. We present two complementary empirical studies that together characterize a systematic *coordination deficit* in current large language model (LLM) agent systems. First, we benchmark emergent coordination in a population of 90,704 autonomous LLM agents and find a pronounced core-periphery role structure (silhouette 0.91), heavy-tailed information cascades ($\alpha = 2.57$), and substantially worse collaborative task outcomes than matched single-agent baselines (Cohen’s $d = -0.88$). Second, using 19,200 agent-scenario pairs across four models, we show that baseline LLMs exhibit systematic *rationality bias* — attenuated herding (0.61 vs. human benchmark 0.70), low loss aversion ($\lambda = 1.12$ vs. 2.25), and near-zero disposition effects — that partly explains why open populations fail to coordinate. Profile-based calibration closes much of this gap (herding $\rightarrow 0.90$; $\lambda \rightarrow 3.00$), providing a direct lever for cooperative system design. Together, the two studies argue that building trustworthy cooperative AI requires both ecological evaluation of what fails at the population level and behavioral calibration of the agent-level priors that cooperation depends upon.

1. Introduction

Cooperative AI — the study of systems in which multiple AI agents coordinate, negotiate, and resolve conflict — rests on a foundational empirical question: do current LLM-based agents actually cooperate, and if not, why? A growing body of work on LLM agents for tool use, reasoning, and long-horizon tasks (Brown et al., 2020; OpenAI, 2023; Schick et al., 2023; Qin et al., 2023) has established individual-

agent competence, and frameworks such as AutoGen (Wu et al., 2023), MetaGPT (Hong et al., 2023), and multi-agent debate (Liang et al., 2023) have shown that structured small-group coordination is achievable under explicit protocols. What remains unknown is how coordination behaves at scale, without scripted roles, and among agents whose behavioral dispositions were shaped by training rather than designed for cooperation.

We argue that answering this question requires two complementary lenses: an *ecological* lens that measures population-level dynamics in realistic agent deployments, and a *behavioral* lens that characterizes the individual-level priors that drive or inhibit cooperative action.

The ecological lens reveals what fails. Drawing on the MoltBook Observatory Archive (Gautam & Riegler, 2026), a public dataset of 2.73M interactions among 90,704 autonomous LLM agents, we find three population-level failure modes: influence concentrates in a tiny active core; information propagates in heavy-tailed, difficult-to-govern cascades; and decentralized collaborative task resolution performs substantially worse than matched single-agent work (Cohen’s $d = -0.88$) (Yee & Koh, 2026a).

The behavioral lens reveals why. Using 19,200 agent-scenario pairs from a controlled measurement study (Yee & Koh, 2026b), we document that baseline LLM agents exhibit systematic rationality bias. They herd less than humans do (0.61 vs. 0.70), anchor less, discount losses less strongly, and underweight social information in belief updating — precisely the priors that human cooperation depends upon. Importantly, these priors are calibratable through profile prompting, opening a path from diagnosis to design.

The two studies have a precise mechanistic connection. Classical collective intelligence research establishes that population-level coordination outcomes depend on both individual behavioral dispositions and network topology (Woolley et al., 2010; Hong & Page, 2004). The ecological failure we document is partly a consequence of agents whose social-behavioral priors sit far from the human cooperative regime; correcting those priors through calibration is therefore not merely a psychometric exercise but a mechanism for improving the population-level outcomes the ecological

¹Yee Collins Research Group. Correspondence to: Brandon Yee <b.yee@ycrg-labs.org>, Pairie Koh <p.koh@ycrg-labs.org>.

study measures.

Our primary contribution is a unified framework that connects ecological evaluation to behavioral calibration as a research agenda for cooperative AI design. The negative result — that open multi-agent coordination imposes a Cohen’s $d = -0.88$ penalty — is presented not as a critique of existing architectures but as a falsifiable benchmark target: coordination protocols claiming cooperative intelligence should be tested against this deficit on ecologically valid tasks.

2. Related Work

Multi-agent LLM systems. LLM-based multi-agent frameworks have demonstrated promising coordination in small structured settings (Wu et al., 2023; Hong et al., 2023; Liang et al., 2023; Park et al., 2023). Emergent communication research shows that agent populations can develop shared protocols through repeated interaction (Lazaridou & Baroni, 2020; Mordatch & Abbeel, 2018), and collective intelligence work establishes that group diversity can improve problem solving over homogeneous groups of high-ability individuals (Hong & Page, 2004). Our ecological study extends this literature to large, unstructured populations where no coordination protocol is externally imposed.

LLMs as behavioral agents. Recent work uses LLMs as simulated economic agents or synthetic survey respondents (Horton, 2023; Aher et al., 2023; Argyle et al., 2023). The behavioral finance literature provides a rich taxonomy of cooperative-relevant priors: herding (Bikhchandani et al., 1992; Anderson & Holt, 1997), overconfidence (Daniel et al., 1998), and extrapolation (Greenwood & Shleifer, 2014) each determine how agents weight private versus social information and how they commit to or revise beliefs under social pressure. We build on this taxonomy to measure cooperative-relevant behavioral priors in LLMs and to calibrate them toward human benchmarks.

Information diffusion and contagion. Classical models distinguish simple from complex contagion (Centola & Macy, 2007) and saturating adoption regimes (Dodds & Watts, 2004). Structural network properties — betweenness, core-periphery organization, brokerage roles — shape both influence and information propagation (Freeman, 1977; Borgatti & Everett, 2000). Our diffusion results instantiate these theoretical regularities in a large LLM-agent population and connect them to governance and safety.

Table 1. Ecological benchmark: each task maps a cooperative-AI concern to an observable proxy and a population-level baseline.

Concern	Proxy	Method	Baseline
Structural governance	Role concentration	Network clustering	Silhouette 0.91
Diffusion safety	Cascade dynamics	Power-law + survival	$\alpha = 2.57$, HR 0.53
Collaboration quality	Task outcomes	Matched comparison	Cohen’s $d = -0.88$

3. Ecological Baselines: Population-Level Coordination

3.1. Dataset and Benchmark Design

The MoltBook Observatory Archive (Gautam & Riegler, 2026), introduced and analyzed in Yee & Koh (2026a), captures a three-week snapshot (January 28 – February 20, 2026) of agents operating on a Reddit-style platform through the open-source OpenClaw framework (Steinberger, 2026). The 90,704 active agents span multiple underlying model families; model identities are not recorded, so our analysis targets platform-level population dynamics rather than architecture-specific comparisons. We observe more than 2.73M interactions, 10,323 extracted information cascades, 8,585,511 adoption events, 60,045 technical discussion threads, and 164 collaborative events.

The benchmark organizes around three failure modes directly relevant to cooperative AI: *structural concentration* (can influence be ungoverned?), *uncontrolled diffusion* (can harmful conventions propagate?), and *collaboration overhead* (does more agents mean worse outcomes?).

3.2. Task 1: Structural Role Differentiation

We construct a directed interaction network over all 90,704 agents, computing in-degree, out-degree, betweenness centrality, clustering coefficient, and PageRank per agent. K-means clustering over standardized features selects $k = 6$ by maximizing the silhouette coefficient (Rousseeuw, 1987), yielding a baseline of 0.91. Bootstrap stability analysis (100 iterations) confirms the clustering structure (mean ARI = 0.960).

The dominant structural feature is extreme core-periphery organization (Borgatti & Everett, 2000): 93.5% of agents fall in a low-activity peripheral cluster. A threshold-based behavioral taxonomy corroborates this: 91.0% of agents are generalists, 8.4% are initiators, 0.5% amplifiers, 0.1% synthesizers, and 0.02% curators. For cooperative AI, this is a governance signal — if synthesis and curation are performed by fewer than 1% of agents, failure or manipulation of that tiny core can have disproportionate effects on the entire population.

3.3. Task 2: Information Diffusion and Cascade Dynamics

We extract cascades via n -gram matching ($n \in \{2, 3, 4, 5\}$) across posts from distinct agents, fitting a power-law distribution following [Clauset et al. \(2009\)](#). Cascade sizes follow a power law with exponent $\alpha = 2.57$ (95% CI [2.54, 2.60]; KS statistic $D = 0.020$). Likelihood-ratio tests favor the power law over exponential ($p < 0.001$) and lognormal ($p = 0.030$) alternatives, consistent with [Newman \(2003\)](#)’s characterization of heavy-tailed social contagion.

Adoption dynamics exhibit saturating contagion: logistic regression yields a positive exposure coefficient (0.328, $p < 0.001$) and a negative quadratic term (-0.0074 , $p < 0.001$), and a Cox proportional hazards model (fitted on 100 sampled cascades; 286,125 observations, 123,590 adoption events) yields a hazard ratio of 0.528 with concordance 0.775. This saturating pattern parallels the complex-contagion regime identified by [Centola & Macy \(2007\)](#): initial exposures raise adoption probability, but repeated contact from the same source progressively reduces adoption hazard. The implication for coordination is that healthy norm propagation requires diverse propagation channels rather than broadcast from a single central agent.

3.4. Task 3: Decentralized Collaboration Benchmarking

We identify 164 multi-agent collaborative events from 60,045 technical threads, defined as threads with at least three distinct agents, at least five substantive exchanges, at least one technical keyword, and a minimum duration of 30 minutes. We evaluate quality using a composite score $Q \in [0, 1]$ weighting code presence (0.30), comment quality (0.30), test inclusion (0.20), and syntax validity (0.20). Each collaborative event is compared against up to five matched single-agent threads controlled for thread length and topic.

The result is stark: collaborative events perform significantly worse than matched single-agent threads ($t = -11.21$, $p = 5.5 \times 10^{-22}$, Cohen’s $d = -0.88$). Only 6.7% of collaborative events achieve $Q \geq 0.5$. Supplementary analysis shows that successful collaborative events ($n = 11$) average 21.0 participants versus 9.4 for unsuccessful ones, suggesting that a critical mass of contributors is associated with quality but is not sufficient on its own. The Cohen’s $d = -0.88$ establishes a hard benchmark: this is the coordination deficit that future cooperative protocols must overcome.

Table 2. Behavioral calibration results for cooperative-relevant parameters. Baseline is the rational-profile value; Max is the highest value induced by a bias-specific profile; Human Benchmark is from meta-analytic or canonical estimates.

Parameter	Baseline	Max	Benchmark	Tier
Loss Aversion (λ)	1.12	3.00	2.25	Strong
Herding Rate	0.61	0.90	0.70	Strong
Extrapolation (θ)	0.44	0.88	0.60	Strong
Anchoring (ρ)	0.61	0.67	0.43	Strong
Overconfidence (coverage)	0.47	0.30	0.65	Directional
Disposition Ratio	0.06	0.21	1.60	Weak
Probability Weight	0.12	0.30	0.35	Moderate
Representativeness	0.15	1.08	1.65	Weak

4. Behavioral Mechanisms: Agent-Level Cooperative Priors

4.1. Measurement Framework

We treat behavioral parameters as latent measurement objects that can be experimentally induced and calibrated within LLMs through profile prompts, following the framework of [Yee & Koh \(2026b\)](#). A profile prompt p functions as an instrument: $p \xrightarrow{\text{induces}} \theta \xrightarrow{\text{determines}} f(I; \theta) \xrightarrow{\text{generates}} d$, where θ is the target behavioral parameter and d is the resulting decision. Valid calibration requires four conditions: monotonicity of the parameter in profile strength, coverage of the human benchmark range, stability across equivalent elicitations, and coherence across experimental contexts (positive correlation between loss aversion measured from gambles and from disposition behavior, negative correlation between overconfidence and herding).

We evaluate eight canonical behavioral parameters — loss aversion (λ), herding rate, extrapolation (θ), anchoring (ρ), overconfidence, disposition ratio, probability weighting, and representativeness — across four LLM variants (GPT-4o, GPT-4o-mini, Claude 3.5 Haiku, Gemini 2.5 Pro) using 19,200 agent-scenario pairs on fully synthetic financial scenarios. All inference applies Bonferroni-Holm correction.

4.2. Rationality Bias and Cooperative Priors

The central finding is systematic rationality bias: across six of eight parameters, baseline LLM behavior sits substantially closer to the rational benchmark than human behavior does (Table 2). This is not a capability failure — it is a distributional one. LLM training corpora frame behavioral biases as cognitive errors, and models internalize this normative orientation.

For cooperative AI, three parameters are directly relevant. *Herding* — the weight agents place on crowd signals versus private information ([Bikhchandani et al., 1992](#); [Anderson & Holt, 1997](#)) — is the most direct cooperative prior. Baseline herding (0.61) falls below the human benchmark (0.70), indicating that LLM agents are systematically over-independent

relative to the social weighting that human coordination relies upon. *Loss aversion* ($\lambda = 1.12$ vs. 2.25) governs commitment: agents with low loss aversion switch positions too readily, undermining the stable commitments that coordination protocols require. *Overconfidence* is the mirror of herding — agents who overestimate their private signal precision (Daniel et al., 1998) discount social information, and LLMs’ baseline overconfidence (coverage 0.47) already exceeds the human benchmark (0.65), amplifying the trust gap.

Profile-based calibration produces large, statistically significant shifts in all three parameters (all $p < 0.001$ after multiple-testing correction, effect sizes $d \geq 0.88$). Herding-prone profiles raise the herding rate from 0.61 to 0.90; loss-averse profiles raise λ from 1.12 to 3.00. Cross-parameter correlations match theoretical predictions: $\text{Corr}(\lambda_{\text{gamble}}, \lambda_{\text{disposition}}) = 0.42$ ($p < 0.001$); $\text{Corr}(\text{overconfidence}, \text{herding}) = -0.31$ ($p < 0.001$). These patterns are inconsistent with undifferentiated stereotype retrieval and confirm that calibration induces economically interpretable parameters.

4.3. External Validity: Calibrated Priors Generate Macro-Level Outcomes

To confirm that calibrated priors carry economic content beyond psychometric fit, we embed the calibrated extrapolation parameter ($\theta = 0.88$) in a simple two-type agent-based asset pricing model. Calibrated extrapolation generates short-horizon momentum of 0.18 and long-horizon reversal of -0.12 , closely matching the empirical stylized facts of Jegadeesh and Titman (0.14 and -0.10 ; 1993). Baseline rational agents generate no momentum. This exercise establishes that calibrated behavioral parameters produce population-level emergent patterns — a result that bridges the behavioral and ecological analyses.

5. From Diagnosis to Design: A Framework for Cooperative AI

The two studies identify a coordination deficit at two levels, but they also point toward a common design agenda. We organize the implications around three mechanisms the cooperative AI community should prioritize.

Mechanism 1: Behavioral calibration as a cooperation lever. The ecological failure is partly a consequence of agents whose social-behavioral priors sit outside the human cooperative regime. If collaboration quality is to approach or exceed the single-agent baseline (the -0.88 deficit), agents participating in collaborative tasks must weight social information more heavily than baseline LLMs do. Profile-based calibration can push herding rates to 0.90 and loss aversion to values that support stable commitments. The design rec-

ommendation is to treat behavioral calibration not as an afterthought but as a core configuration step before deploying agents in multi-agent settings: agents intended to cooperate should be profiled for appropriate social weighting.

Mechanism 2: Structural diversity through designed topology. The core-periphery finding (93.5% peripheral agents; $< 1\%$ curators and synthesizers) suggests that open populations default to structural inequality that concentrates both influence and vulnerability. Since agent network position predicts role differentiation, multi-agent system designers should explicitly engineer structural diversity — including coordination brokers and inter-community bridging roles — rather than assuming useful specialization will emerge spontaneously. Provenance and verification mechanisms, applied differentially to high-centrality agents, can leverage the structural concentration rather than simply lamenting it.

Mechanism 3: Ecological benchmarking as a reporting standard. The three-task benchmark suite — role concentration, cascade dynamics, and collaboration quality — provides a platform-agnostic template for evaluating cooperative systems under realistic interaction conditions. We recommend that future cooperative AI systems report these ecological metrics alongside capability benchmarks. A protocol that improves average task success while worsening the collaboration deficit is not a genuine advance in cooperative intelligence. The $d = -0.88$ baseline is a concrete, replicable target: any coordination protocol claiming cooperative intelligence should demonstrate improvement over this deficit on ecologically valid tasks.

These three mechanisms are complementary rather than competing. Behavioral calibration improves individual-agent cooperative priors; structural design creates the topological conditions for those priors to produce population-level coordination; ecological benchmarking verifies that population-level outcomes improve in ways that match the theoretical predictions of the behavioral and structural interventions.

6. Limitations and Conclusion

The ecological study has three important limits. The three-week observation window may not capture long-run coordination dynamics. Model identities are not recorded, preventing attribution to specific architectures. N-gram cascade detection misses semantic adoption, and the matched collaboration comparison cannot fully remove task-selection effects.

The behavioral study has complementary limits. Calibration succeeds for cognitive-computational aspects of behavioral priors — herding, loss aversion, extrapolation, anchoring — but fails for parameters whose human manifestation relies

on emotional intensity, social pressure, or real monetary stakes (disposition effects and representativeness remain well below human benchmarks). Behavioral profiles calibrated in isolated scenarios may not persist under adversarial multi-agent interaction.

Despite these limits, the studies jointly characterize a systematic and quantifiable coordination deficit in current LLM agent systems, trace it to behavioral priors that are identifiable and calibratable, and provide a benchmark framework that makes future progress falsifiable. The appropriate response to the $d = -0.88$ result is not pessimism about multi-agent AI but a clear engineering agenda: calibrate cooperative priors, design for structural diversity, and evaluate ecologically. Safe and beneficial cooperative AI requires all three.

Impact Statement

This paper presents work whose goal is to advance the scientific understanding of multi-agent LLM coordination and its failures. The primary societal benefit is methodological: the ecological benchmark and behavioral calibration framework provide tools for auditing and improving cooperative AI systems before they are deployed in consequential settings. We identify a coordination deficit that, if left unaddressed, could cause multi-agent deployments to underperform single-agent baselines in real-world tasks — a risk that this work makes measurable and therefore actionable. We are not aware of specific harms that must be individually highlighted here beyond those that motivate the work.

References

- Aher, G., Arriaga, R. I., and Kalai, A. T. Using large language models to simulate multiple humans and replicate human subject studies. *International Conference on Machine Learning*, pp. 337–371, 2023.
- Anderson, L. R. and Holt, C. A. Information cascades in the laboratory. *American Economic Review*, 87(5):847–862, 1997.
- Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., and Wingate, D. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3):337–351, 2023.
- Bikhchandani, S., Hirshleifer, D., and Welch, I. A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5): 992–1026, 1992.
- Borgatti, S. P. and Everett, M. G. Models of core/periphery structures. *Social Networks*, 21(4):375–395, 2000.
- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33: 1877–1901, 2020.
- Centola, D. and Macy, M. Complex contagions and the weakness of long ties. *American Journal of Sociology*, 113(3):702–734, 2007.
- Clauset, A., Shalizi, C. R., and Newman, M. E. Power-law distributions in empirical data. *SIAM Review*, 51(4): 661–703, 2009.
- Daniel, K., Hirshleifer, D., and Subrahmanyam, A. Investor psychology and security market under-and overreactions. *Journal of Finance*, 53(6):1839–1885, 1998.
- Dodds, P. S. and Watts, D. J. Universal behavior in a generalized model of contagion. *Physical Review Letters*, 92(21):218701, 2004.
- Freeman, L. C. A set of measures of centrality based on betweenness. *Sociometry*, 40(1):35–41, 1977.
- Gautam, S. and Riegler, M. A. Moltbook observatory archive, 2026. URL <https://huggingface.co/datasets/SimulaMet/moltbook-observatory-archive>. Incremental export of MoltBook observatory database as date-partitioned Parquet files.
- Greenwood, R. and Shleifer, A. Expectations of returns and expected returns. *Review of Financial Studies*, 27(3): 714–746, 2014.
- Hong, L. and Page, S. E. Groups of diverse problem solvers can outperform groups of high-ability problem solvers. *Proceedings of the National Academy of Sciences*, 101(46):16385–16389, 2004.
- Hong, S., Zheng, X., Chen, J., Cheng, Y., Zhang, J., Wang, C., Wang, Z., Yau, S. K. S., Lin, Z., Zhou, L., et al. Metagpt: Meta programming for a multi-agent collaborative framework. *arXiv preprint arXiv:2308.00352*, 2023.
- Horton, J. J. Large language models as simulated economic agents: What can we learn from homo silicus? *National Bureau of Economic Research Working Paper*, (31122), 2023.
- Jegadeesh, N. and Titman, S. Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1):65–91, 1993.
- Lazaridou, A. and Baroni, M. Emergent multi-agent communication in the deep learning era. *arXiv preprint arXiv:2006.02419*, 2020.
- Liang, T., He, Z., Jiao, W., Wang, X., Wang, Y., Wang, R., Yang, Y., Tu, Z., and Shi, S. Encouraging divergent thinking in large language models through multi-agent debate. *arXiv preprint arXiv:2305.19118*, 2023.
- Mordatch, I. and Abbeel, P. Emergence of grounded compositional language in multi-agent populations. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 2018.
- Newman, M. E. The structure and function of complex networks. *SIAM Review*, 45(2):167–256, 2003.
- OpenAI. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- Park, J. S., O’Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., and Bernstein, M. S. Generative agents: Interactive simulacra of human behavior. In *UIST*, 2023.
- Qin, Y., Liang, S., Ye, Y., Zhu, K., Yan, L., Lu, Y., Lin, Y., Cong, X., Tang, X., Qian, B., et al. Toolllm: Facilitating large language models to master 16000+ real-world apis. *arXiv preprint arXiv:2307.16789*, 2023.
- Rousseeuw, P. J. Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987.
- Schick, T., Dwivedi-Yu, J., Dessì, R., Raileanu, R., Lomeli, M., Zettlemoyer, L., Cancedda, N., and Scialom, T. Toolformer: Language models can teach themselves to use tools. *Advances in Neural Information Processing Systems*, 36, 2023.

- Steinberger, P. Openclaw: An open-source framework for autonomous ai agents. <https://github.com/steipete/openclaw>, 2026. Accessed: 2026-01-31.
- Woolley, A. W., Chabris, C. F., Pentland, A., Hashmi, N., and Malone, T. W. Evidence for a collective intelligence factor in the performance of human groups. *Science*, 330 (6004):686–688, 2010.
- Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, S., Liu, J., et al. Autogen: Enabling next-gen llm applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*, 2023.
- Yee, B. and Koh, P. Benchmarking emergent coordination in large-scale llm populations: An evaluation framework on the moltbook archive, 2026a. URL <https://arxiv.org/abs/2603.03555>.
- Yee, B. and Koh, P. Calibrating behavioral parameters with large language models, 2026b. URL <https://arxiv.org/abs/2602.01022>.

A. Supplementary Details: Ecological Study

A.1. Role Evaluation

Network features (in-degree, out-degree, betweenness centrality, clustering coefficient, PageRank) were standardized to zero mean and unit variance before k-means clustering. Bootstrap stability analysis (100 iterations with replacement) yielded mean silhouette 0.913 (SD 0.004; 95% CI [0.899, 0.917]) and mean ARI 0.960 (SD 0.037; 95% CI [0.854, 0.998]). K-means outperformed Gaussian Mixture Models (silhouette 0.914 vs. 0.805; ARI between solutions = 0.49).

A.2. Diffusion Analysis

Power-law estimation used maximum likelihood following [Clauset et al. \(2009\)](#). The lower cutoff $x_{\min} = 104$; KS statistic $D = 0.020$. Likelihood-ratio comparisons: power law over exponential (log-LR = 6109.6, $p < 0.001$); power law over lognormal (log-LR = 0.22, $p = 0.030$); comparison with truncated power law inconclusive ($p = 0.997$). Cox proportional hazards: coefficient -0.638 , HR 0.528, SE 0.003, $z = -186.2$, $p < 0.001$; Schoenfeld global test $\chi^2 = 2.1$ ($p = 0.15$), confirming no violation of the proportional hazards assumption.

A.3. Collaboration Analysis

Quality score components (successful vs. unsuccessful events): code presence 0.82 vs. 0.14; comment quality 0.65 vs. 0.28; test inclusion 0.45 vs. 0.02; syntax validity 0.73 vs. 0.08. Successful collaborations averaged 21.0 participants (mean duration 3.2 hours) versus 9.4 participants (1.7 hours) for unsuccessful events.

B. Supplementary Details: Behavioral Study

B.1. Calibration Effects

Profile-based calibration effects (rational $\rightarrow$ biased profile): loss aversion $1.12 \rightarrow 3.00$ ($d = 1.82$, $p < 0.001$); herding rate $0.61 \rightarrow 0.90$ ($d = 2.15$, $p < 0.001$); extrapolation $0.44 \rightarrow 0.88$ ($d = 1.95$, $p < 0.001$); anchoring $0.61 \rightarrow 0.67$ ($d = 1.43$, $p < 0.001$). All effects significant after Bonferroni-Holm correction.

B.2. Model Heterogeneity

Baseline herding rates by model: GPT-4o 0.56; GPT-4o-mini 0.58; Gemini 2.5 Pro 0.78; Claude 3.5 Haiku 0.54. All models show baseline parameters below human benchmarks for most biases. Loss aversion, herding, and extrapolation show coefficient of variation < 0.20 across models, indicating stable calibration. Disposition effects and representativeness show $CV > 0.40$, indicating unreliable measurement for these parameters.

B.3. Asset Pricing Validation

The agent-based model uses a single risky asset with fundamental value following a random walk, with rational traders ($\theta = 0$) and extrapolative traders ($\theta > 0$) each submitting demand proportional to expected return scaled by risk aversion $\gamma = 2$. LLM-calibrated extrapolation ($\theta = 0.88$) generates short-horizon momentum 0.18 and long-horizon reversal -0.12 ; human-calibrated ($\theta = 0.60$) generates 0.12 and -0.08 ; baseline rational generates no momentum. Empirical target from [Jegadeesh & Titman \(1993\)](#): 0.14 and -0.10 . Results are robust to heterogeneous extrapolation (continuous $\theta \sim \text{Uniform}[0, 0.88]$), proportional trading costs ($c = 0.1\%$), and stochastic news arrivals.